Out put of my analysis

This homework involves data analysis, pre-processing, outlier detection, and handling missing values,and Testing the values to make a conclusion and visualization.
After running the analysis below is my analysis

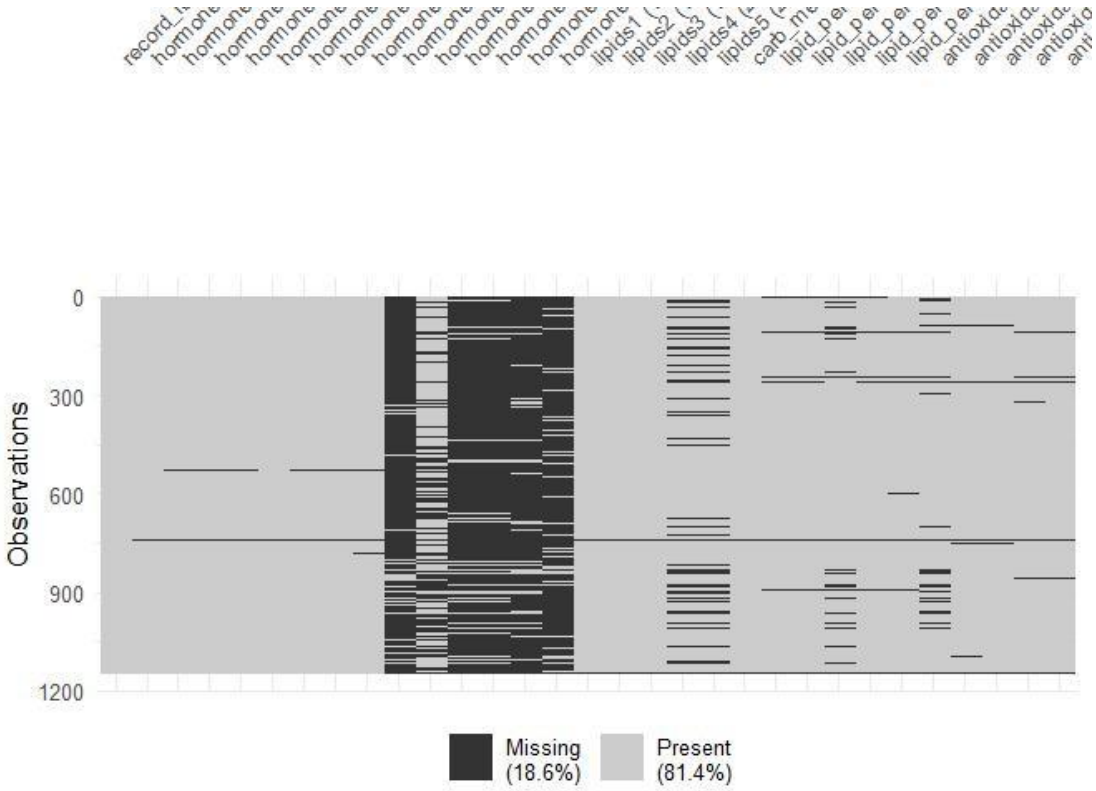

Figure 1 (Missing data pattern plot)

Here in the figure above, we can see that the grey areas are the areas of present data (81.4%) and the black area show the missing values (18.6%).

Data_Manipulation_Process.R × mcar_result ×

Filter

	statistic	df	p.value	missing.patterns
1	1809.34	1102	0	54

Figure 2 (Little MCAR Test)

From the above figure , we can see that the calculated test value and df is 1102 which is the degree of freedom with the p value of 0 which is less than 0.005 therefore we can conclude that missing values are not random because

p-value = 0.0000 ≤ 0.05

So we Reject H_0 . Therefore: The data is NOT MCAR (it is either MAR or MNAR). So I considered more sophisticated missing data handling method thereby using multiple imputation as the right choice.

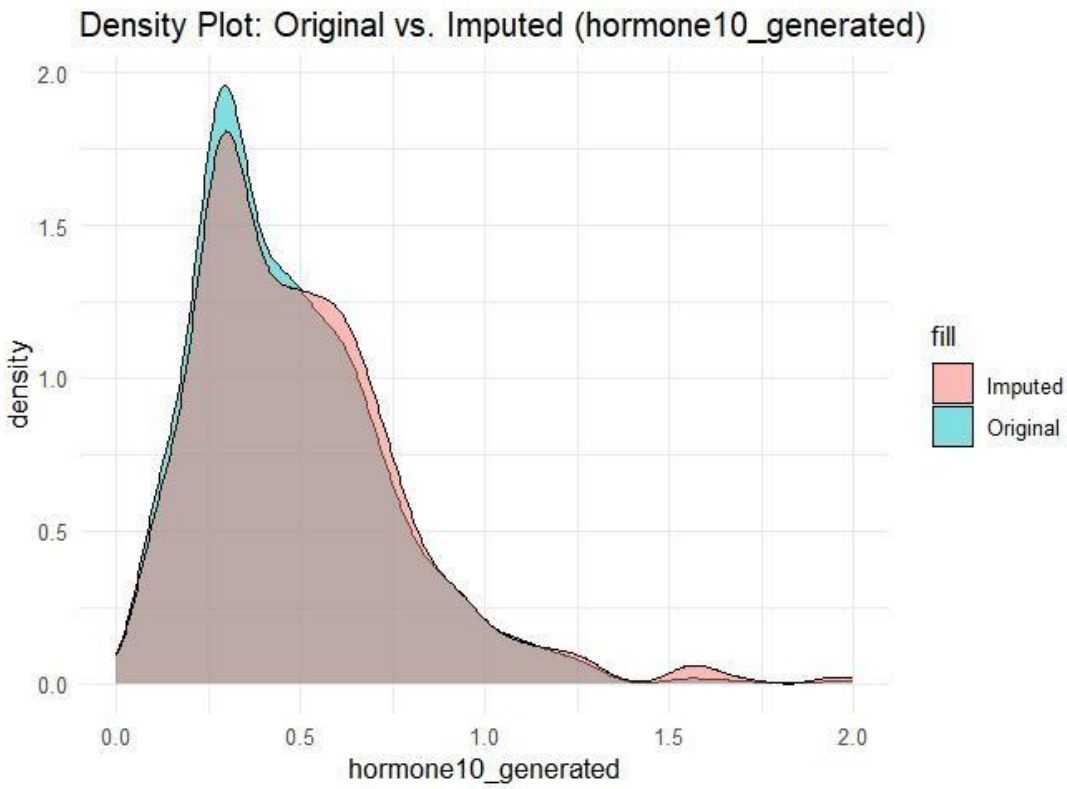

Figure 3 : Hormone Density Plot

From the diagram we can see the Density Plot comparing the distribution of the variable hormone10_generated before and after filling missing values. The blue area shows the distribution of the original data that was already present while the Brown area shows the distribution after imputation filling in the missing values.Therefore we can say that the shapes are

almost identical, which means the method used to fill the gaps worked very well because it preserved the original pattern and spread of the data without changing it unnaturally.

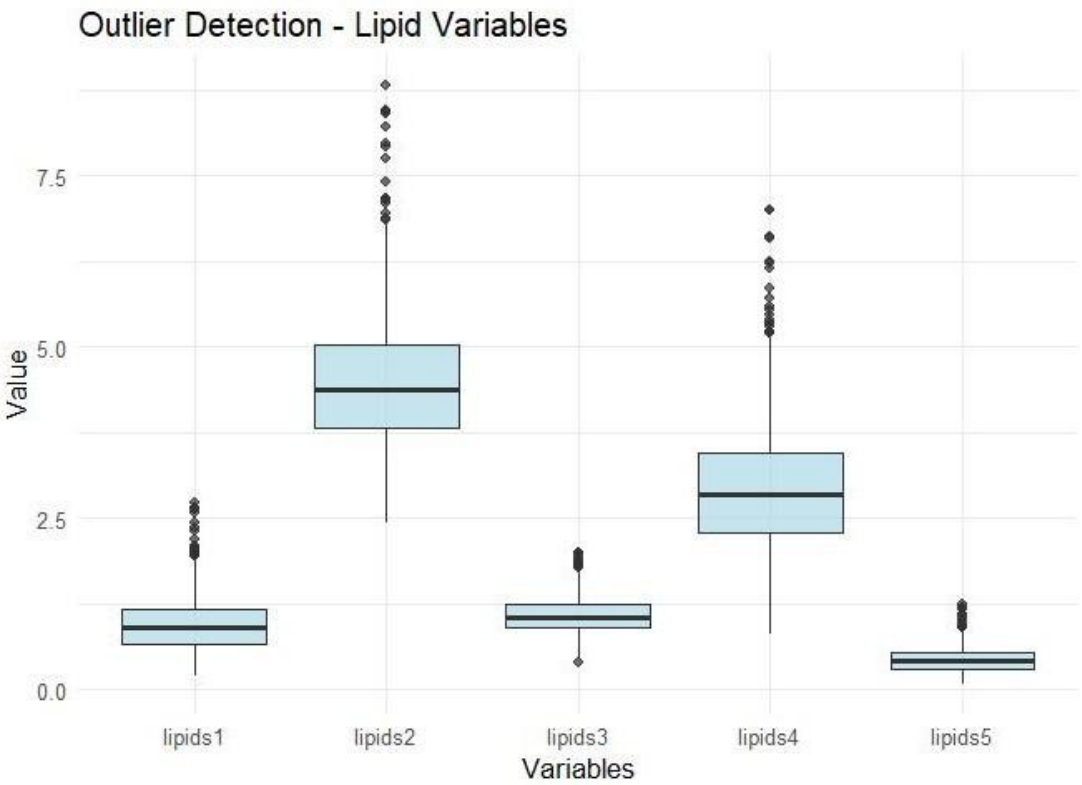

Figure 4 : Outlier Detection boxplot

From the figure we see that it compares the distribution of values for 5 different lipid measurements (lipids1 to lipids5). The small circles above and below the boxes are outliers values that are unusually high or low compared to the rest.We observed that lipids2 and lipids4 have the most and highest outliers, meaning they contain the most extreme values.

```
lof_data        872 obs. of 26 variables
  $ record_id      : int  1 2 3 4 6 8 9 10 12 15 ...
  $ hormone1       : num  0.5 1.9 1.2 3.8 1.6 1.6 0.42 2.7 1 2.…
  $ hormone2       : int  314 196 311 409 418 309 415 450 198 5…
  $ hormone3       : num  9.9 6.8 1.2 3.2 3.8 2.7 5.9 7 9 6.9 .…
  $ hormone4       : num  4.2 6.8 3.2 2.7 7.2 1.6 9.1 4.5 5.2 4…
  $ hormone5       : num  322.7 190.9 153.7 225.7 99.9 ...
  $ hormone6       : num  166.1 41 140.8 93.4 24.7 ...
  $ hormone7       : num  5.7 6.8 5.4 12.7 1.6 9.9 7.5 4 4.2 5.…
```

Figure 5 : Local factor calculation

From figure 5 we can see that 872 observations and 26 varaibles so therefore we can see that the data is clean , complete and prepared for outlier detection.

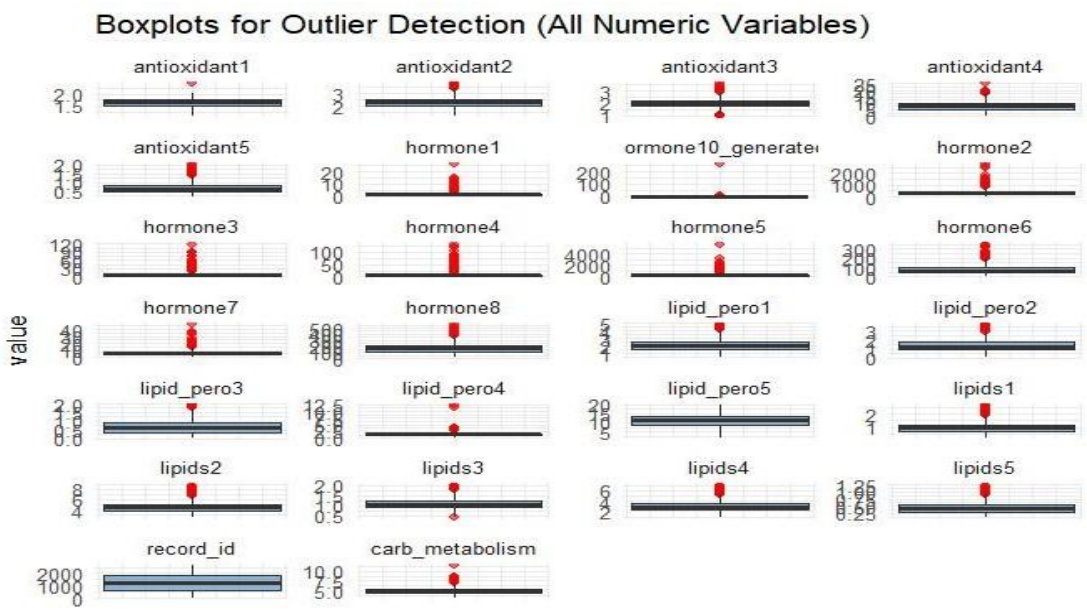

Figure 6 : Boxplots for Outlier Detection

From the figure above, we can visually check every numerical column in the dataset for unusual or extreme values and it shows the distributions for variables like antioxidant1-5 , hormone1-8 , lipid_pero1-5 , lipids1-5 etc.The red dots represent outliers values that stand far away from the normal range of the data. And we can see that almost every variable has some outliers, especially the hormone and lipid measurements, indicating many extreme values exist across the data.

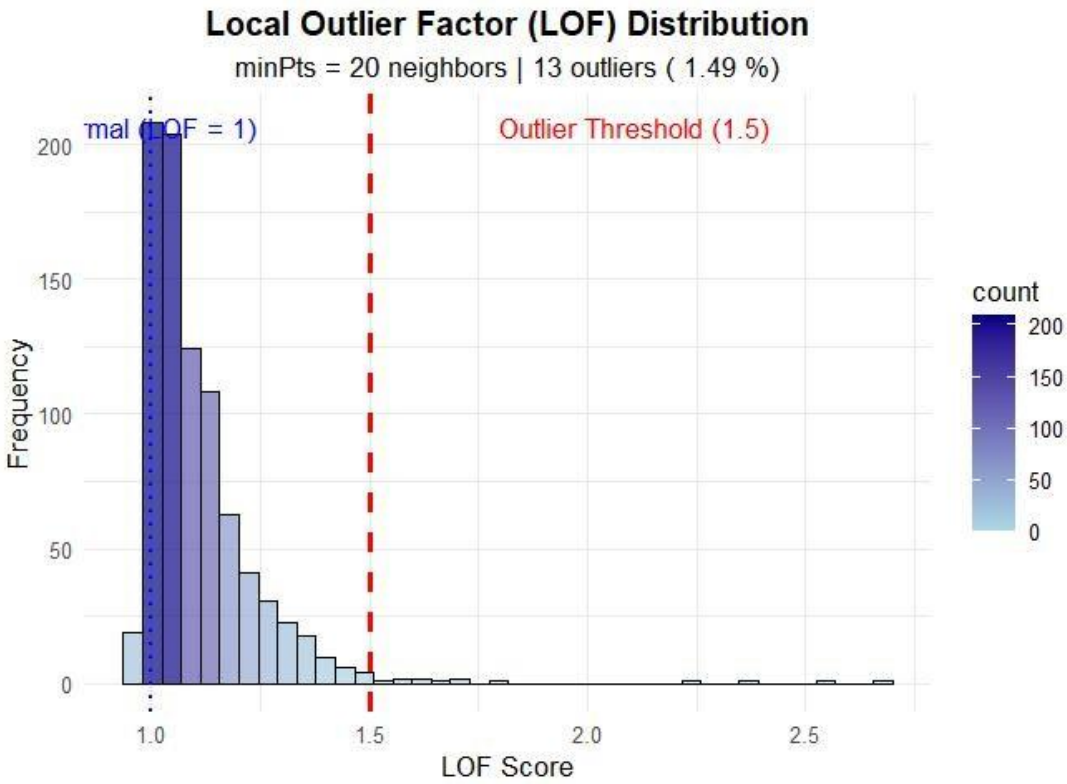

Figure 7 : H istogram for Outlier Detection.

From the figure above, the histogram shows the distribution of LOF scores. Most points are on the left with scores close to 1, meaning they are normal. While the red dashed line shows the threshold at 1.5 and any point with a score above it is considered an outlier. So therefore, using 20 neighbors, we found 13 outliers, which is only 1.49% of the total data.It therefore means that the vast majority of your data is consistent and normal, with very few extreme or unusual value

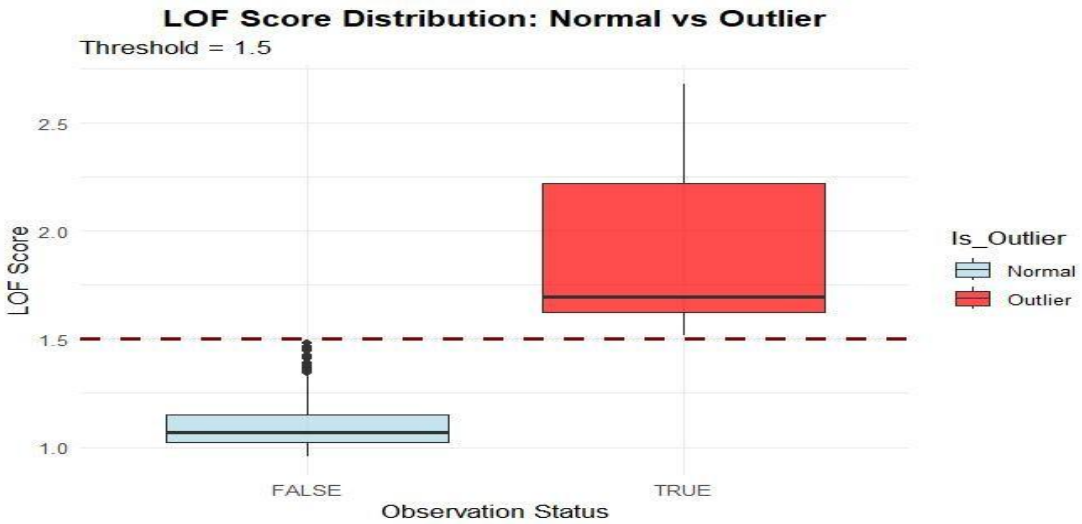

Figure 8 : Boxplots comparing LOF scores

From the figure above, we can see the boxplot comparing LOF scores between normal data and outliers. Left side (FALSE) shows scores for normal data points. They are all below the red dashed line (threshold = 1.5), meaning they are consistent and not unusual. While the right side (TRUE) which shows scores for the detected outliers. From the figure we can see they are all clearly above the threshold, confirming they are different from the rest.Therefore due to the separation is very clear, therefore it means the method worked well and correctly identified which points are outliers.

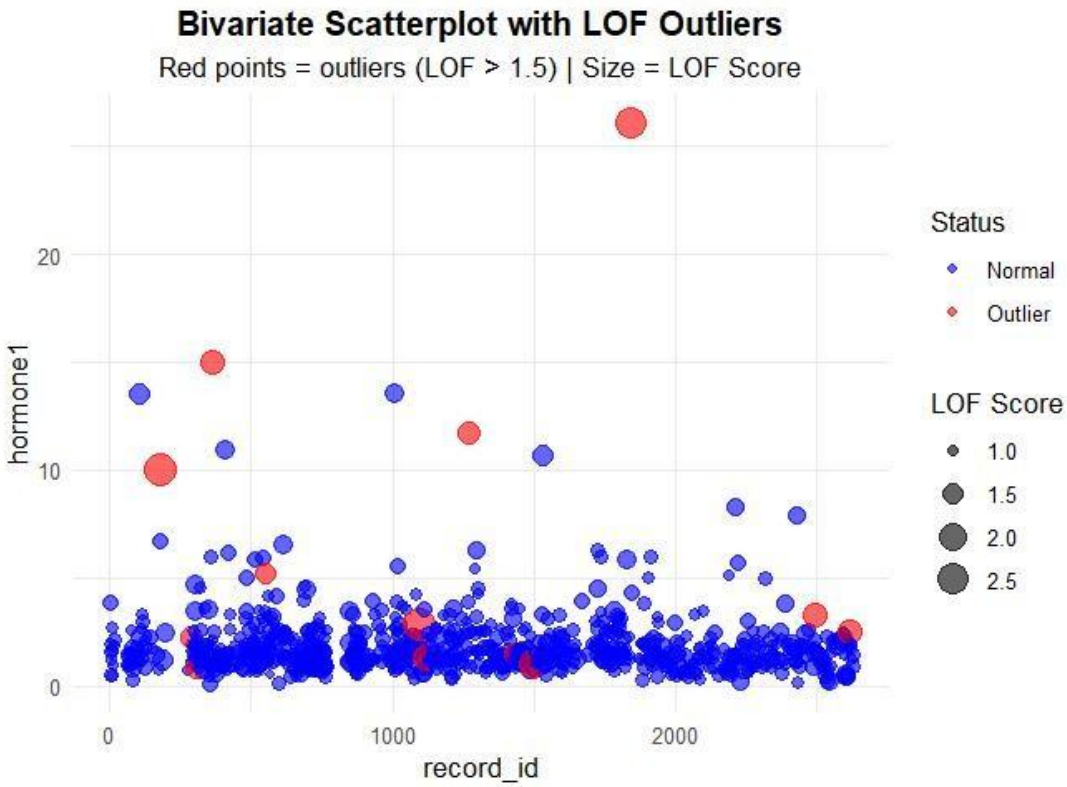

Figure 9: Bivariate Scatterolot

From the figure above, we can see that the scatterplot displays hormone1 levels across record IDs, where blue points represent normal observations and red points highlight outliers detected by the Local Outlier Factor (LOF) algorithm with scores above 1.5. We can see that the majority of data points form a dense cluster near the bottom (hormone1 values mostly 0–5), while several red points stand out significantly higher, indicating anomalous hormone spikes, with larger red dots showing stronger outliers.

Conclusion

We can conclude that the missing data mechanism was identified as MAR or MNAR (not MCAR). Also, PMM imputation was successfully completed with high quality with multivariate outliers were identified and lastly some visualizations were generated for result communication.